

Performance & Scale Readiness Report — BErozgar

1. Executive Summary

The BErozgar (V0.1.0) platform currently exhibits a baseline architectural posture typical of early-stage Vite + React single-page applications (SPAs) backed by Node.js and PostgreSQL. While the modern stack (Vite, Fastify, Prisma) provides excellent developer experience and baseline execution speed ^{[1][5]}, the production configuration lacks critical performance hardening required for campus-wide deployment. Because the application relies entirely on client-side rendering (CSR) with no Server-Side Rendering (SSR), users will experience a blank screen until the JavaScript bundle is fully downloaded, parsed, and executed ^{[3][8]}.

The readiness verdict is **NOT READY** for a full campus launch. The architecture will fail under concurrent load spikes (e.g., a mass email blast announcing placements) due to database connection exhaustion and monolithic asset delivery.

Top 3 Critical Risks:
Prisma Connection Pool Exhaustion: Without an external pooler like PgBouncer, concurrent Fastify requests will rapidly exhaust PostgreSQL's default `max_connections` (typically **100**), leading to cascading timeout failures ^{[10][13]}.
Monolithic JavaScript Bundle: Vite's default production build without explicit `manualChunks` configuration will ship a massive, render-blocking JS payload, severely degrading Largest Contentful Paint (LCP) on campus Wi-Fi or mobile networks ^{[3][8]}.
N+1 Query Inefficiencies: Unoptimized Prisma queries fetching relational data (e.g., jobs + companies + application counts) will flood the database with sequential queries, multiplying latency ^[11].

2. Observed Metrics with Actual Numbers

The following metrics represent the current V0.1.0 performance posture. Estimated values are derived from standard Vite/React SPA baselines running behind unoptimized Nginx/Docker infrastructure.

Metric	Value	Source	Google Web Vitals Threshold	Status
Time to First Byte (TTFB)	~150–400ms	Estimated	< 800ms	Pass (Marginal)
First Contentful Paint (FCP)	~1.8–2.5s	Estimated	< 1.8s	Needs Improvement
Largest Contentful Paint (LCP)	~2.8–4.2s	Estimated	< 2.5s	Fail
Total Blocking Time (TBT)	~300–600ms	Estimated	< 200ms	Fail
Speed Index	~3.5–5.0s	Estimated	< 3.4s	Fail

Lighthouse Performance Score	45–65 / 100	Estimated	90+	Fail
Nginx Version Signature	Visible	Measured	N/A	Security Risk
Directory Listing (/assets/)	403 Forbidden	Measured	N/A	Pass

3. Bundle Analysis

- **Vite Production Build Characteristics:**Vite utilizes Rollup for production builds, generating hashed filenames in the `/assets` directory for cache busting^{[2][5]}. However, without explicit configuration, Vite bundles all application code and third-party dependencies into a single monolithic JavaScript file^{[3][8]}.
- **Code Splitting Assessment:**By default, Vite does not aggressively split vendor chunks for React SPAs. The likely bundle composition is a single `index-[hash].js` containing React, ReactDOM, routing libraries, and all application logic. This forces the browser to download the entire application before rendering the login page^[3].
- **Tree Shaking:**Vite/Rollup performs statement-level tree shaking automatically, removing unused exports^[2]. However, large dependencies (like UI component libraries or date utilities) will still bloat the bundle if imported incorrectly^[6].
- **Estimated Bundle Sizes:**At V0.1.0, a typical Vite+React app yields an unoptimized bundle of **500KB – 1.2MB**(gzipped)^[8].
- **Render-Blocking Risk:**Because this is a pure SPA with no SSR, the render-blocking risk is **100%**. The HTML shell is empty; the browser must download, parse, and execute the entire JS bundle before the user sees any UI^[5].
- **Recommendation:**Implement route-level code splitting using `React.lazy()` and `Suspense`^{[4][7]}. Configure `manualChunks` in `vite.config.ts` to separate stable vendor libraries (React, Fastify clients) from application code, reducing the initial payload by up to **95%**^{[3][4][8]}.

4. Infrastructure Assessment

Component	Current Status	Risk Level	Recommendation
Nginx SPA Config	Unknown	High	Ensure <code>try_files \$uri \$uri/ /index.html;</code> is configured. Without this, direct navigation to deep links (e.g., <code>/dashboard</code>) will return a 404 Not Found ^[14] .

Compression	Unknown	High	Enable Brotli (nginx-module-brotli) and Gzip. Uncompressed JS/CSS will increase payload sizes by 300–400% , crippling mobile load times ^{[15][19]} .
HTTP/2	Unknown	Medium	Configure listen 443 ssl http2;. HTTP/2 multiplexing is critical for SPAs to download multiple split chunks concurrently without TCP connection limits ^{[17][19]} .
CDN	None Observed	Medium	Implement Cloudflare or AWS CloudFront. Serving static assets directly from the origin server increases latency for users outside the immediate data center region ^[16] .
Caching Headers	Unknown	High	Apply Cache-Control: public, max-age=31536000, immutable to the /assets/ location block. Hashed filenames guarantee cache safety ^[16] .
Docker Networking	Standard Bridge	Low	Standard bridge networking adds minor latency. Ensure Fastify connects to PostgreSQL via internal Docker DNS, not external IP routing.

Security Headers	Unconfirmed	High	Implement the 6 critical headers: Strict-Transport-Security (HSTS), X-Content-Type-Options, X-Frame-Options, X-XSS-Protection, Referrer-Policy, and Content-Security-Policy (CSP) ^{[18][21]} .
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5. Prisma / PostgreSQL N+1 Risk Analysis

Prisma's ORM abstraction easily masks N+1 query inefficiencies. The following patterns will degrade performance under load ^[11]:

1. **Job listings page (Jobs + Company + App Count):***Pattern:* Fetching jobs, then looping to fetch the company and application count for each job.*Fix:* Use Prisma `include` with `_count`.*Code:*

```
prisma.job.findMany({ include: { company: true, _count: { select: { applications: true } } } })
```
2. **Student dashboard (Applications + Job + Company):***Pattern:* Fetching applications, then fetching the job, then the company.*Fix:* Nested includes.

```
include: { job: { include: { company: true } } }
```
3. **Admin panel (Students + Application History):***Pattern:* Fetching all users, then querying applications per user.*Fix:*

```
prisma.user.findMany({ where: { role: 'STUDENT' }, include: { applications: true } })
```
4. **Job search with filters (Department, Batch, CTC):***Pattern:* Fetching all jobs and filtering in memory (Node.js).*Fix:* Push filtering to the database using Prisma's `where` clauses. Ensure composite indexes exist for the filtered columns.
5. **Notification system (Unread count per page load):***Pattern:* Fetching all notification records to count them.*Fix:* Use

```
prisma.notification.count({ where: { user_id: userId, is_read: false } })
```

instead of `findMany`.

6. PostgreSQL Indexing Requirements

Without these indexes, PostgreSQL will perform sequential table scans, causing CPU spikes and query timeouts as the database grows ^[11].

- **users table:** `CREATE UNIQUE INDEX idx_users_email ON users(email);` (Optimizes auth lookups). `CREATE INDEX idx_users_role_batch ON users(role, batch_year);` (Optimizes admin filtering).

- **job table:** `CREATE INDEX idx_jobs_company ON jobs(company_id);` (Optimizes relational joins).
`CREATE INDEX idx_jobs_status_created ON jobs(status, created_at DESC);` (Optimizes the main job feed sorting).
- **applications table:**
`CREATE INDEX idx_apps_student ON applications(student_id);` (Optimizes student dashboard).
`CREATE UNIQUE INDEX idx_apps_student_job ON applications(student_id, job_id);` (Prevents duplicate applications and optimizes lookup).
- **notifications table:**
`CREATE INDEX idx_notif_user_read ON notifications(user_id, is_read);` (Optimizes the global unread count query).

7. Scaling Bottlenecks — Ranked by Impact

Tier 1: 0 → 1,000 concurrent users

- **First Bottleneck:** PostgreSQL Connection Pool Exhaustion.
- **Failure Mode:** Prisma creates a connection pool per Fastify instance. If scaled horizontally, or under high concurrency, the default PostgreSQL `max_connections` (**100**) will be exceeded, resulting in
Timed out fetching a new connection from the connection pool errors ^{[9][10][13]}.
- **Mitigation:** Tune Prisma's `connection_limit` in the `DATABASE_URL` and ensure a singleton PrismaClient instance is used ^{[9][13]}.

Tier 2: 1,000 → 10,000 concurrent users

- **First Bottleneck:** Database Memory and Connection Overhead.
- **Failure Mode:** Even with tuned Prisma pools, the sheer volume of connections will consume excessive PostgreSQL RAM (each connection uses **~5–10MB**) ^[12].
- **Mitigation:** Implement **PgBouncer** in transaction mode to multiplex thousands of client connections onto a small number of actual PostgreSQL connections ^{[12][13]}.

Tier 3: 10,000 → 100,000 concurrent users

- **First Bottleneck:** Database Read IOPS and Node.js Event Loop.
- **Failure Mode:** Repeated identical queries (e.g., fetching the same active job listings) will max out database CPU. Fastify will struggle to serialize massive JSON payloads.
- **Mitigation:** Introduce Redis for caching read-heavy, infrequently changing data (job listings, company profiles). Offload all static asset delivery to a CDN.

8. Top 10 Optimization Actions

Priority	Action	Effort	Impact	Category	Details
1	PgBouncer Integration	M	High	Database	Deploy PgBouncer in transaction mode to prevent PostgreSQL connection exhaustion under load ^{[12][13]} .
2	Vite Code Splitting	M	High	Frontend	Implement React.lazy() for routes and manualChunks for vendor libraries to shrink the initial JS payload ^{[3][4][8]} .
3	Nginx HTTP/2 & Brotli	S	High	Infra	Enable http2 and nginx-module-brotli to multiplex requests and compress text assets by up to 80% ^{[15][17][19]} .
4	Prisma Query Tuning	M	High	Backend	Replace sequential queries with Prisma include and _count to eliminate N+1 database calls ^[11] .
5	PostgreSQL Indexing	S	High	Database	Apply the 8 critical indexes outlined in Section 6 to prevent sequential table scans.

6	Nginx Caching Headers	S	Med	Infra	Add Cache-Control: immutablefor /assets/to leverage browser caching for returning students ^[16] .
7	Security Headers	S	Med	Security	Apply HSTS, CSP, and X-Frame-Options via Nginx to prevent XSS and clickjacking ^{[18][21]} .
8	Fastify Rate Limiting	S	Med	Security	Install @fastify/rate-limit to protect login and application submission endpoints from brute-force/DoS
9	CDN Setup	S	Med	Infra	Route traffic through Cloudflare to cache static assets at the edge and provide baseline DDoS
10	Image Optimization	M	Low	Frontend	protection ^[16] . Ensure company logos and profile pictures are served in WebP format with lazy loading (loading="lazy").

- **HSTS (Strict-Transport-Security):** Implementing HSTS (`max-age=31536000; includeSubDomains; preload`) not only secures the application but improves performance by forcing the browser to default to HTTPS, eliminating the latency of a 301 redirect from HTTP to HTTPS on subsequent visits ^{[15][18]}.
- **Content Security Policy (CSP):** A strict CSP prevents malicious script execution but requires careful configuration. If misconfigured, it will block legitimate inline scripts or styles, breaking the SPA ^[18].
- **Auth Token Overhead:** The application uses HMAC_SHA256. If this implies stateful session tokens stored in the database, every authenticated request requires a database lookup, adding latency. If it implies signed JWTs, the token size increases HTTP header payload size, but eliminates the database read.
- **Rate Limiting:** Without rate limiting, a simple script can overwhelm the Fastify server or exhaust the Prisma connection pool, causing a denial of service. Rate limiting must be enforced at the Nginx level (`limit_req_zone`) or via Fastify ^{[16][20]}.

10. Overall Performance Readiness Verdict

- **Readiness Score:** 4 / 10
- **Production ready for:** Closed Alpha testing, QA swarms, and limited user groups (< 50 concurrent users).
- **NOT ready for:** Campus-wide launch, mass email campaigns, or high-concurrency placement drives.
- **Minimum actions before go-live:** Configure Vite `manualChunks` and route-level lazy loading ^{[3][8]}. Deploy PgBouncer to protect PostgreSQL ^[12]. Apply database indexes to `users`, `job`, and `applications`. Enable Nginx Brotli compression, HTTP/2, and caching headers ^{[15][17]}.
- **Timeline estimate to reach production-grade performance:** 2 to 3 weeks for a single senior engineer to implement, test, and verify the infrastructure and code-level optimizations.

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